

SAT UZBEKISTAN

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1.

In triangle ABC , the measure of angle A is 54° and the measure of angle B is 90° . What is the measure, in degrees, of angle C ?

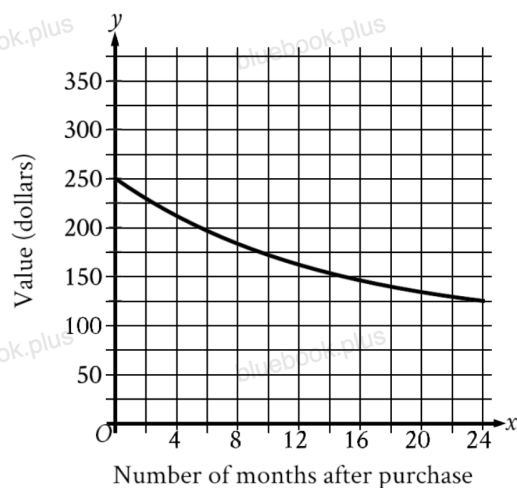
(A) 36

(B) 54

(C) 90

(D) 180

2.



The graph shown gives the estimated value, in dollars, of a mountain bike as a function of the number of months since it was purchased. What is the best interpretation of the y -intercept of the graph in this context?

(A) The estimated value of the bike was \$250 when it was purchased.

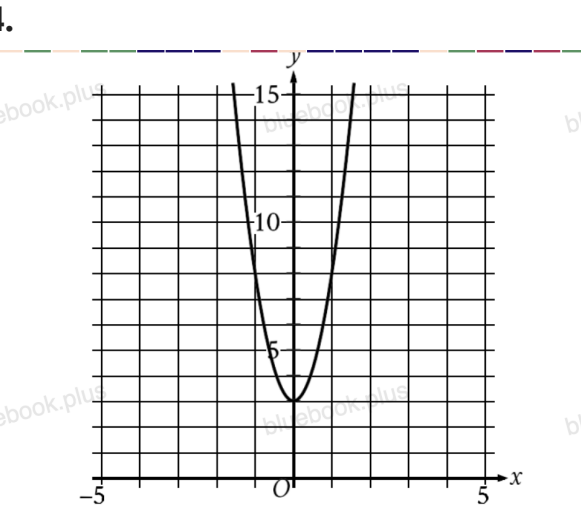
(B) The estimated value of the bike 24 months after it was purchased was \$250.

(C) The estimated value of the bike had decreased by \$250 in the 24 months after it was purchased.

(D) The estimated value of the bike decreased by approximately 2.50% each year after it was purchased.

3. A certain rectangular flag is produced in several different sizes. The flag is designed so that its length is always **1.59** times its width. Which equation gives the area A , in square feet, of this flag if it has a width of w feet?

- ☐ A $A=(1.59w)(w)$
- ☐ B $A=(\frac{1.59}{w})(w)$
- ☐ C $A=(w-1.59)(w)$
- ☐ D $A=(w+1.59)(w)$



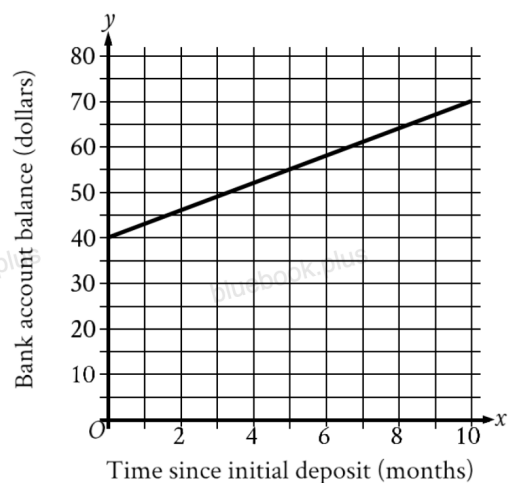
The graph of $y = f(x)$ is shown. Which equation defines f ?

- ☐ A $f(x)= 5(x + 1)^2 + 2$
- ☐ B $f(x)= 5(x + 1)^2$
- ☐ C $f(x)= 5x^2+3$
- ☐ D $f(x)= 5x^2$

5. One minute after the start of an experiment, the volume of a mixture is 95% of its volume at the start of the experiment. Which expression **best** represents the volume of the mixture one minute after the start of the experiment, where v is its volume at the start of the experiment?

- ☐ A $0.095v$
- ☐ B $0.95v$
- ☐ C $9.5v$
- ☐ D $95v$

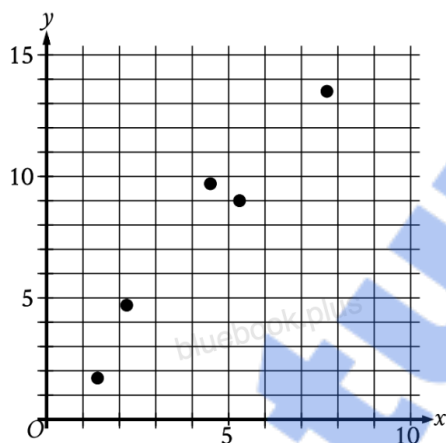
6.



A bank account was opened with an initial deposit. Over the next several months, regular deposits were made into this account, and there were no withdrawals made during this time. The graph of the function f shown, where $y = f(x)$, estimates the account balance, in dollars, in this bank account x months since the initial deposit. To the nearest whole dollar, what is the amount of the initial deposit estimated by the graph?

7.

The scatterplot shows the relationship between two variables, x and y .



Which of the following equations is the most appropriate linear model for the data shown?

(A) $y = -1.8x - 0.2$

(B) $y = -0.2x - 1.8$

(C) $y = 0.2x + 1.8$

(D) $y = 1.80x + 0.2$

8.

Antwon's hair will grow at a constant rate from a length of **3** inches to a length of **6** inches. The equation $0.2t + 3 = 6$ describes this situation, where t is the number of months it will take for Antwon's hair to reach a length of **6** inches. Which of the following is the best interpretation of **0.2** in this context?

(A) The time, in months, it will take Antwon's hair to grow **1** inch

(B) The time, in months, it will take Antwon's hair to reach a length of **6** inches

(C) The length, in inches, Antwon's hair will grow each month

(D) The length, in inches, Antwon's hair will grow in **6** months

9.

$$39x + 78 = r(3x + 6)$$

In the given equation, r is a positive integer. If this equation has more than one solution, what is the value of r ?

10.

For the function f , for every increase of $\frac{1}{2}$ in the value of x , the value of $f(x)$ increases by a factor of c , where c is a constant. Which of the following forms of function f displays the value of c as the base or coefficient?

A) $f(x) = 56(2)^{6x}$

B) $f(x) = 56(8)^{2x}$

C) $f(x) = 56(64)^x$

D) $f(x) = 56(4,096)^{\frac{x}{2}}$

11.

A plant nursery classifies a perennial flower species as tall if its typical height when fully grown is more than 100 centimeters. Each of the following inequalities represents the possible heights h , in centimeters, for a specific perennial flower species when fully grown. Which inequality represents the possible heights h , in centimeters, for a tall perennial flower species?

(A) $11 < h < 86$

(B) $48 < h < 89$

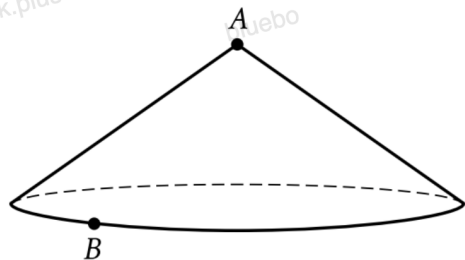
(C) $77 < h < 100$

(D) $114 < h < 154$

12.

For a certain type of rope, the equation $y = 900ax^2$, where a is a constant, gives the estimated breaking strength y , in pounds, of a rope with a circumference of x inches. Based on this equation, if a rope of this type has a circumference of **1.25** inches, it has an estimated breaking strength of **1,968.75** pounds. What is the estimated breaking strength, in pounds, of a rope of this type that has a circumference of **4.50** inches?

13.



Note: Figure not drawn to scale.

For the right circular cone shown, B is a point on the circumference of the base, and the length of segment AB (not shown) is **78** centimeters. If the height of the cone is **39** centimeters and the volume of the cone is $k\pi$ cubic centimeters, what is the value of k ?

14.

$$z(w) = (0.835)^{3w}$$

The function z is defined by

the equation shown. The

value of $z(w)$ decreases by

$p\%$ for each increase by **1** in

the value of w . Which of the

following is closest to the

value of p ?

(A) 41.8

(B) 16.5

(C) 0.835

(D) 0.165

15.

$$-2|x - 7| = -8x$$

What are all possible solutions to the given equation?

(A) $-\frac{7}{3}$

(B) $\frac{7}{5}$

(C) $-\frac{7}{3}$ and $\frac{7}{5}$

(D) $\frac{7}{5}$ and $\frac{7}{3}$

16.

The table shown summarizes the distribution of 65 employees based on the number of complete years they have worked at a certain company.

Number of complete years at the company	Number of employees
0 to 3	7
4 to 6	17
7 to 9	14
10 to 12	9
13 to 15	5
16 to 18	10
19 to 21	3

How many of the 65 employees have worked at the company for 16 or more complete years?

(A) 10

(B) 13

(C) 52

(D) 65

17.

	Live north of Center St.	Live south of Center St.	Total
Less than 45 years old	21	11	32
At least 45 years old	14	89	103
Total	35	100	135

The table summarizes members of a local organization by age and whether they live north St. If a member of the organization is selected at random, what is the probability that the selected member is at least 45 years old?

(A) $\frac{32}{135}$

(B) $\frac{35}{135}$

(C) $\frac{100}{135}$

(D) $\frac{103}{135}$

18.

The perimeter of an isosceles triangle is 66 feet. Each of the two congruent sides of the triangle has a length of 19 feet. What is the length, in feet, of the third side?

(A) 19

(B) 22

(C) 28

(D) 33

19.

$$y = 2x^2 - 13x + 24$$

$$y = 3x + a$$

In the given system of equations, a is a constant. The graphs of the equations in the given system intersect at exactly one point, (x, y) , in the xy -plane. What is the value of x ?

(A) -8

(B) -4

(C) 4

(D) 8

20.

The force F , in newtons, acting upon an object is the product of the mass of the object, in kilograms (kg), and the acceleration a , in meters per second squared (m/s^2), of the object. Which equation represents the force F , in newtons, acting on an object with a mass of 33 kg and an acceleration of $a \text{ m/s}^2$?

(A) $F = 33a$

(B) $F = 33 + a$

(C) $F = 33 - a$

(D) $F = a - 33$

21.

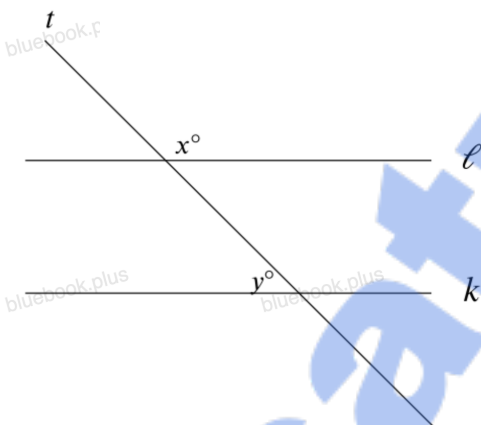
The number of bacteria in a population doubles every hour. The function $f(t) = 34,000(2)^t$ gives the number of bacteria in the population, where t represents the time, in hours, after the initial measurement. Which is the best interpretation of 34,000 in this context?

- ☐ A The number of bacteria in the population at the initial measurement
- ☐ B The number of hours it takes for the number of bacteria in the population to double
- ☐ C The number of bacteria in the population 1 hour after the initial measurement
- ☐ D The number of times the number of bacteria in the population doubles in 1 hour

22.

A certain blue whale calf weighed 5,900 pounds when it was born, and its weight increased at an average rate of 250 pounds per day for the first n days after birth, where n is an integer. If the calf's weight was greater than 8,650 pounds but less than 9,650 pounds n days after birth, what is one possible value of n ?

23.



☐ A $y < 80$

☐ B $y > 80$

☐ C $x + y < 180$

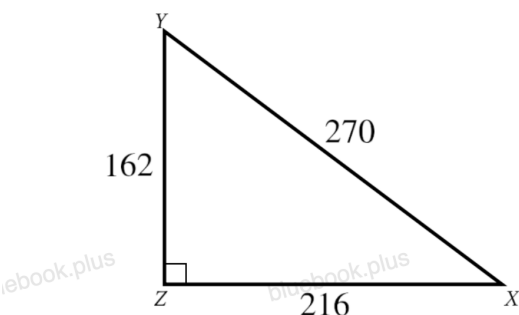
☐ D $x + y > 180$

In the figure shown, lines l and k are parallel and line t intersects both lines. If $x > 100$, following must be true?

24.

A circle in the xy -plane has its center at $(7, 2)$ and a radius with a length of 6. The equation for this circle can be written in the form $(x - h)^2 + (y - k)^2 = b$, where h , k , and b are constants. What is the value of h ?

25.



Note: Figure not drawn to scale.

What is the value of $\tan X$ in the triangle shown?

26.

Square A has side lengths that are 98 times the side lengths of square B . The area of square A is k times the area of square B . What is the value of k ?

☐ A 49

☐ B 98

☐ C 196

☐ D 9,604

27.

For the exponential function f , the value of $f(3)$ is k , where k is a constant. Which of the following equivalent forms of the function f shows the value of k as the coefficient or the base?

A) $f(x) = 3(4)(4)^{x+1}$

B) $f(x) = 3,072(4)(4)^{x-4}$

C) $f(x) = 12(4)(4)^x$

D) $f(x) = 768(4)(4)^{x-3}$

28.

At the start of a stock trading competition, John's and Julia's stock portfolios were of equal value. At the end of the stock trading competition, John's portfolio had increased by 330%. Julia's portfolio has increased by 2,480%. At the end of the competition Julia's portfolio is worth $p\%$ more than John's portfolio, what is the value of p ?

29.

$$\frac{3}{5}x + \frac{7}{4}y = a$$

$$\frac{7}{11}x + by = c$$

In the given system of equations, a , b , and c are constants. The system has infinitely many solutions. What is the value of $\frac{a}{c}$?

30.

In the given pair of equations a and b are constants. The graph of this pair of equations in the xy -plane is a pair of perpendicular lines. Which of the following pairs of equations also represents a pair of perpendicular lines?

$$13x + 17y = 10$$

$$ax + by = 10$$

- A) $39x + 17y = 10$
 $ax - 3by = 10$
- B) $52x + 34y = 10$
 $2ax + 4by = 10$
- C) $65x + 17y = 10$
 $5ax + by = 10$
- D) $13x - 17y = 10$
 $ax + by = 10$

31.

$$f(x) = (x + a)(x + b)$$

For the given function $f(x)$, where a and b are integer constants, $f(-21) > 0$, $f(-15) > 0$, and $f(-18) < 0$, what is one possible value of $a + b$?

32.

$$f(x) = (x - a)(x - b)$$

The function f is defined above, where a and b are integer constants. If $f(1) > 0$, $f(4) < 0$, and $f(7) > 0$, what is one possible value of $a + b$?

33.

$(x + 7)$ is a factor of the quadratic function $y = f(x)$, and $x = c$ is a positive zero of the function. If the minimum of the function is $(\frac{63}{8}, k)$, what is the value of c ?

34.

For two acute angles, $\angle A$ and $\angle B$, $\cos A = \sin B$. The measures, in degrees, of $\angle A$ and $\angle B$ are $13x + 10$ and $48 + 3x$, respectively. What is the value of x ?

35.

A baseball coach charges each athlete \$150 for the first month and \$95 for each additional month. Which of the following functions correctly represents the total revenue $R(m)$, in dollars, for m months and n athletes, where m and n are positive integers?

- A) $R(m) = 95mn + 55$
- B) $R(m) = 95mn + 150$
- C) $R(m) = 95mn + 55n$
- D) $R(m) = 95mn + 150n$

36.

$$y = -3x^2 - 58$$

$$y = px - 46$$

In the given system of equations, p is a constant. The graphs of the equations in the given system intersect at exactly one point, (x, y) in the xy -plane. Which of the following could be the value of x ?

- A) -2
- B) -4
- C) 12
- D) 16

37.

$$f(x) = -7x + b$$

For the given function $f(x)$, which of the following gives the equation for the function $f(x) + 20$, given that the point $(-4, 0)$ lies on the graph of $f(x) - 20$?

- A) $f(x) = -7x - 28$
- B) $f(x) = -7x - 8$
- C) $f(x) = -7x + 8$
- D) $f(x) = -7x + 12$

38.

The positive number a is 480% of the number b , and a is 70% of the number c . If c is $p\%$ of b , which of the following is closest to the value of p ?

- A) 267
- B) 336
- C) 550
- D) 685

39.

$$f(x) = a^x - b$$

The exponential function given above passes through points $(c, 8)$ and $(2c, 350)$. What is a possible value of b ?

40.

Cylinder	Volume	Surface Area
A	$1,296\pi$	$k\pi$
B	$162,000\pi$	$j\pi$

The table gives the volume of two similar cylinders. If the radius of cylinder A is 9 cm. What is the value of $j - k$?

Volume = $\pi r^2 h$
 Surface area = $2\pi r^2 + 2\pi r h$

41.

$$g(x) = ab^{\frac{x}{n}}$$

In the given function, $g(6) = 11$ and $g(9) = 704$, what is the value of $g(11)$?

42.

A rectangle is inscribed in a circle such that the length of the diagonal of the rectangle is twice the length of its shortest side. The area of the circle is $5,929\pi$. What is the area, in square units, of the rectangle?

- A) $77\sqrt{2}$
- B) $77\sqrt{3}$
- C) $5,929\sqrt{2}$
- D) $5,929\sqrt{3}$

43.

In triangle ABC and triangle DEF , sides BC and EF each have a side length of 23 inches, and angles C and F each have an angle measure of 43° . Which of the following pieces of information is(are) sufficient to prove whether triangle ABC is congruent to triangle DEF ?

- I. The measures of angles D and E are equal.
- II. The lengths of sides AB and DE are equal.
- III. The measures of angles A and D are equal.

- A) I is sufficient, but II and III are not.
- B) II is sufficient, but I and III are not.
- C) II and III are sufficient, but I is not.
- D) III is sufficient, but I and II are not.

44.

The speed of object K is 9,525% of the speed of object J , and the speed of object K is 0.025% of the speed of object L . If the speed of object L is $x\%$ of the speed of object J , what is the value of $\frac{x}{1,000}$?